

Project Idea: Field-Granular Schema-Annotated Trust for Offline-First Synchronization

The Problem

- **Vulnerability in Current Systems:** Offline-first distributed databases (like PouchDB and CouchDB) default to a Last-Write-Wins (LWW) strategy to resolve synchronization conflicts. This relies entirely on timestamps, allowing any compromised node to forge a timestamp and ensure its malicious update is accepted as the authoritative version.
- **The Granularity Mismatch:** While previous models intercept syncs using Weighted Majority Voting (WMV) and global Beta trust models, they apply a single trust threshold to the entire document. This causes two critical failures:
 - **Over-trusting (Reputation Laundering):** Malicious devices build high global trust through minor, honest interactions with low-stakes fields, then use that accumulated trust to corrupt critical data within the same document.
 - **Under-trusting (System Rigidity):** If the global threshold is set conservatively high to protect sensitive clinical or financial fields, legitimate newer users are blocked from making routine, low-risk administrative updates, leading to data loss.

The Proposed Solution This project proposes building a Node.js middleware synchronization interceptor that shifts from document-level trust to a **field-granular, schema-annotated trust framework**.

- **Schema-Annotated Metadata ("Compliance-as-Code"):** We will extend standard JSON Schema to allow domain experts to directly embed risk parameters into the data model. Custom keywords like `tau` (required cumulative trust threshold, $\tau_{\{f\}}$) and `lambda` (global-to-field reputation inheritance, $\lambda_{\{f\}}$) will define the trust landscape for every single attribute.
- **Hierarchical Bayesian Trust Engine:** Instead of starting from zero for every new field, the system uses a hierarchical posterior model where field-level trust estimates can inherit from the user's global Beta distribution. The $\lambda_{\{f\}}$ weight determines this transfer: critical fields ignore global trust ($\lambda_{\{f\}} \rightarrow 0$) to isolate risk, while low-risk fields leverage it completely ($\lambda_{\{f\}} = 1$) to solve the cold-start problem.
- **Per-Field Weighted Majority Voting (WMV):** The interceptor isolates conflicts using dot-notation pathing and executes log-odds weighted voting independently for each field.

System Architecture & Key Outcomes

- **Partial Commits:** The most significant feature is the "partial commit". The merge assembler can accept legitimate updates to some fields (e.g., a display name) while rejecting untrusted modifications to others (e.g., medication dosage) within the exact same document synchronization event.

- **Performance & Accuracy Metrics:** Based on preliminary simulations, this granular approach pushes malicious update rejection rates to >99% for critical risk tiers. Simultaneously, it reduces the False Positive Rate (FPR) for legitimate administrative updates from 3-5% down to <1%.
 - **Overhead:** The architecture maintains high efficiency, keeping synchronization latency under 80ms and trust metadata memory usage below 200 bytes per user, making it ideal for IoT and mobile environments.
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